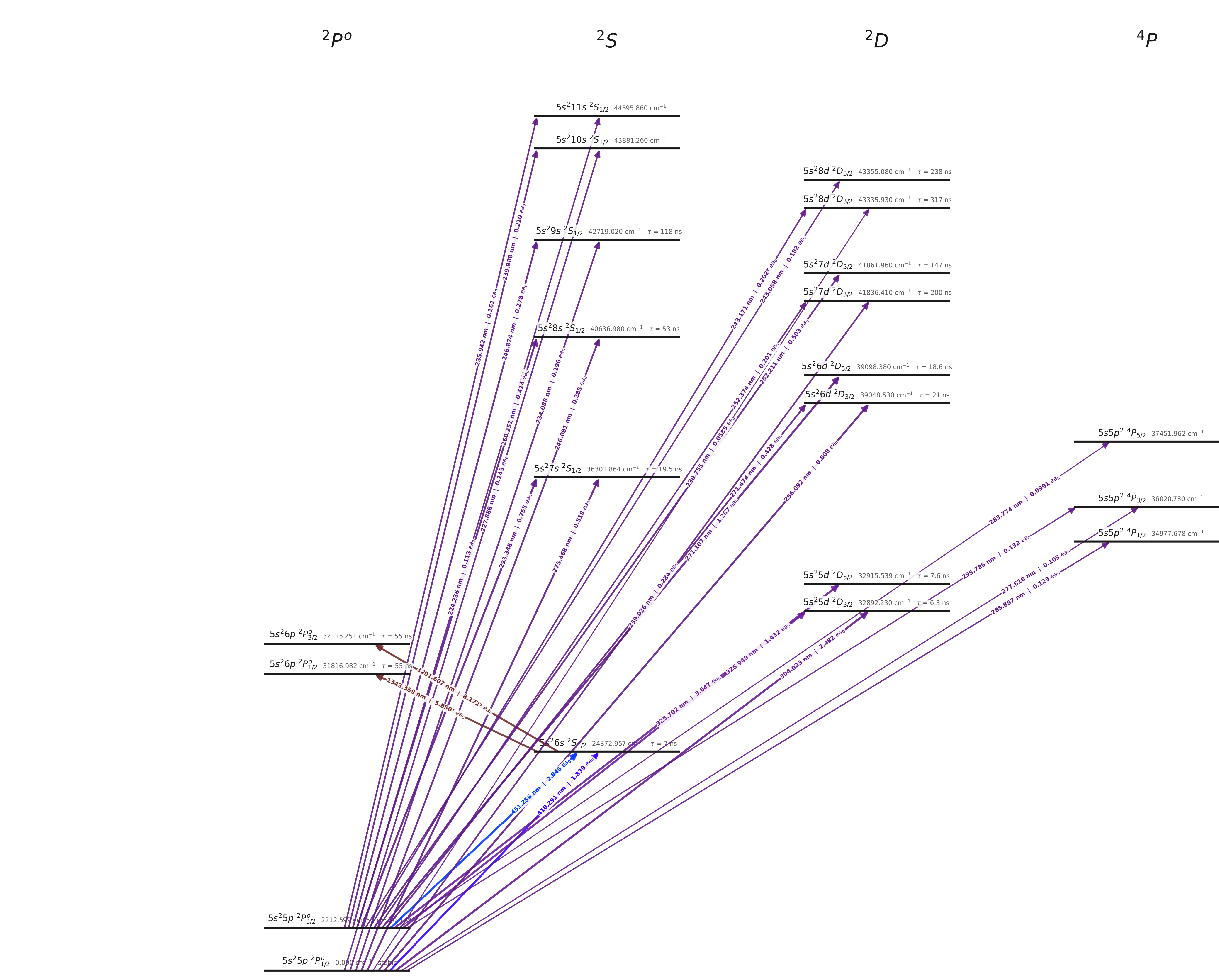


Indium

In (In I). Strongest classified transitions below 2 μm (NIST): 21 levels, 30 lines.

In⁺ ionisation limit 46670.11 cm⁻¹ = 5.78636 eV (214.270 nm)



Key transitions

transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s ² 5p ² P _{1/2} ^o – 5s ² 8s ² S _{1/2}	246.0813	246.0068	1218.2660	0.285	5.510e+06	3.003	29.2	246.0 nm line used for collinear resonance ionization spectroscopy
5s ² 5p ² P _{3/2} ^o – 5s ² 9s ² S _{1/2}	246.8744	246.7998	1214.3519	0.278	5.210e+06	1.349	61.5	246.8 nm line used for collinear resonance ionization spectroscopy
5s ² 5p ² P _{3/2} ^o – 5s ² 5d ² D _{5/2}	325.7017	325.6078	920.4510	3.647	1.300e+08	20.94	100	325.6 nm laser-cooling / MOT transition 5P3/2
5s ² 5p ² P _{1/2} ^o – 5s ² 6s ² S _{1/2}	410.2908	410.1750	730.6829	1.839	4.960e+07	22.74	35.7	410.2 nm resonance line
5s ² 5p ² P _{3/2} ^o – 5s ² 6s ² S _{1/2}	451.2562	451.1297	664.3508	2.846	8.930e+07	22.74	64.3	451.1 nm line
5s ² 6s ² S _{1/2} – 5s ² 6p ² P _{3/2} ^o	1291.6069	1291.2536	232.1081	8.172*	1.570e+07	2.894	100	1291 nm second step of two-step spectroscopy
5s ² 6s ² S _{1/2} – 5s ² 6p ² P _{1/2} ^o	1343.3593	1342.9920	223.1663	5.850*	1.430e+07	2.894	100	1343 nm second step of two-step spectroscopy

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹¹⁵In hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5s ² 5p ² <i>P</i> _{1/2} ^o	2281.9501(4)	-	F=4: -6275.36 F=5: +5134.39
5s ² 5p ² <i>P</i> _{3/2} ^o	242.1647(3)	449.545(3)	F=3: -1791.82 F=4: -1122.86 F=5: -5.69 F=6: +1747.00
5s ² 6s ² <i>S</i> _{1/2}	1685.3(6)	-	F=4: -4634.57 F=5: +3791.93
5s ² 6p ² <i>P</i> _{1/2} ^o	251.40(10)	-	F=4: -691.35 F=5: +565.65
5s ² 6p ² <i>P</i> _{3/2} ^o	79.33(7)	62.5(5)	F=3: -625.83 F=4: -350.17 F=5: +33.46 F=6: +551.10
5s ² 5d ² <i>D</i> _{3/2}	-64.0(10)	41(20)	F=3: +546.79 F=4: +263.46 F=5: -65.08 F=6: -421.75
5s ² 5d ² <i>D</i> _{5/2}	151.2(9)	33(20)	F=2: -2063.88 F=3: -1620.17 F=4: -1024.72 F=5: -274.23 F=6: +635.45 F=7: +1709.25
5s ² 7s ² <i>S</i> _{1/2}	541.0(3)	-	F=4: -1487.75 F=5: +1217.25
5s5p ² 4 <i>P</i> _{5/2}	3696.0(10)	-660(30)	F=2: -51122.50 F=3: -39836.50 F=4: -24865.50 F=5: -6275.50 F=6: +15851.00 F=7: +41415.00
5s ² 8s ² <i>S</i> _{1/2}	243.85(24)	-	F=4: -670.59 F=5: +548.66
5s ² 9s ² <i>S</i> _{1/2}	131.1(10)	-	F=4: -360.44 F=5: +294.91

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |*U||er||*J'*'*| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5s ² 5p ² <i>P</i> _{1/2} ^o – 5s ² 8s ² <i>S</i> _{1/2}	246.081	115-113	268.0(20)
5s ² 5p ² <i>P</i> _{3/2} ^o – 5s ² 9s ² <i>S</i> _{1/2}	246.874	115-113	-270.0(20)
5s ² 5p ² <i>P</i> _{1/2} ^o – 5s ² 6s ² <i>S</i> _{1/2}	410.291	115-113	-260(3)
5s ² 5p ² <i>P</i> _{3/2} ^o – 5s ² 6s ² <i>S</i> _{1/2}	451.256	113-115	255.4(5)

Shift = ν(first isotope) – ν(second isotope).